

Efficiency and abatement costs of energy-related CO₂ emissions in China: A slacks-based efficiency measure

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ABSTRACT

This paper uses nonparametric efficiency analysis technique to estimate the energy efficiency, potential emission reductions and marginal abatement costs of energy-related CO₂ emissions in China. We employ a non-radial slacks-based data envelopment analysis (DEA) model for estimating the potential reductions and efficiency of CO₂ emissions for China. The dual model of the slacks-based DEA model is then used to estimate the marginal abatement costs of CO₂ emissions. An empirical study based on China's panel data (2001–2010) is carried out and some policy implications are also discussed.

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1. Introduction

China's reform and open policy has allowed the country to achieve remarkable progress in economic and social development. Over the past three decades from 1979 to 2009, China's nominal GDP has increased by approximately 83-fold. However, Chinese economic growth was mainly fueled by energy-intensive heavy industries and infrastructure construction, which resulted in a huge amount of energy consumption. In 2000, China's total energy consumption was only about half of that in the United States. However, according to BP [1], in 2010 China's energy consumption was higher than that of the United States by 150 million tons of oil equivalent (TOE). Clearly, China has already overtaken the United States and become the world's largest energy consumer.

China's energy consumption has been overwhelmingly dominated by fossil fuel, which produced significant carbon dioxide (CO₂) emissions. Since 2007, China has been the largest contributor of CO₂ emissions to the world. Scale-oriented economic development in China has given rise to the problems associated with high energy consumption and serious environmental pollution. China's 12th Five-Year Plan (2011–2015) [2] sought to promote a “green, low-carbon development paradigm”. This plan seems to be the first official effort to include a commitment to the gradual introduction

of market mechanisms to control carbon pollution. Recently, the Chinese government announced several new energy and CO₂ emission targets by 2015 with 2010 as the base year, e.g., increasing the proportion of non-fossil fuels in energy consumption to 11.4%, reducing energy consumption per unit of gross domestic product (GDP) by 16%, and reducing CO₂ emissions per unit of GDP by 17% by 2015.

Additionally, the carbon controlling mechanism stipulated in the 12th Five-Year Plan includes the establishment of low-carbon product standards, the improvement of the statistical accounting systems for greenhouse gas emissions, and the promotion of step-by-step implementation of a carbon emission trading system (ETS).¹ The initial plan was to establish carbon emissions trading schemes in several pilot regions including Beijing, Chongqing, Shanghai, Tianjin, Hubei, and Guangdong. Therefore, in an effort to come into compliance with a “low-carbon economy” regime, it is worthwhile to estimate the limits of CO₂ emission reduction potential and the abatement cost of CO₂ emissions in China, which may provide useful information for the Chinese government to bring ETS into the market in the near future. As far as we know, almost

¹ The EU ETS launched in 2005 is the first and biggest international model for greenhouse gas emission trading based on “cap and trade” principle. The study by Zhang and Wei [3] provides an excellent overview of EU ETS with emphases on its operating mechanisms and economic effects. Although China has not established its carbon emission trading scheme, some researchers, e.g. Cong and Wei [4], have devoted to investigate the potential impacts of carbon emission trading on the economy.

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no studies have analyzed the efficiency and abatement costs of CO₂ emissions simultaneously in China from a productive efficiency perspective.² In addition, the most previous studies use the radial efficiency measures that cannot capture all the technical inefficiency in energy use and CO₂ emissions. It is therefore the purpose of this study to fill in the gap by employing a slacks-based efficiency measure for estimating the emission reduction potentials and marginal abatement costs of CO₂ emission in China, the world's largest energy consumer and CO₂ emissions producer.

The remainder of the paper is structured as follows: Section 2 reviews the related literatures. Section 3 explains the methodology of this study and Section 4 presents the results of the analysis and the discussion. Section 5 concludes the paper.

2. Literature review

Distance function³ approach is widely employed to estimate energy/environmental efficiency and the shadow prices of pollutants because it can provide a total-factor efficiency indicator and can incorporate undesirable outputs into the model. One can estimate the distance function in at least two different ways, i.e. the nonparametric data envelopment analysis (DEA) approach and the parametric approach. A comparison between different methods for benchmarking building energy performance can be found in [8]. The DEA estimation is based on the construction of a piecewise linear combination of all observed outputs and inputs, which relies on mathematical programming for the computation of efficiency scores. A major advantage of the DEA approach is that it does not need the imposition of a functional form on the underlying technology. Therefore, DEA would provide an estimate of efficiency easily. The parametric approach, on the other hand, requires the adoption of a functional form (usually the translog function) for the distance function. It has the advantage of providing an estimated parametric representation of the technology that is everywhere differentiable and easy to manipulate algebraically. Therefore, the parametric method is usually used to estimate shadow prices of bad outputs and the curvature or substitutability along the frontier. For the parametric approach used in shadow price and substitutability, one can refer to [7] for details.

In the line of the parametric method, Färe et al. [9] first propose to employ the Shephard output distance function to derive the shadow prices of pollutants. Hailu and Veeman [10] utilize the Shephard input distance function to estimate the production technical efficiency and shadow prices for pollutants. Zhou et al. [11] use the Shephard energy distance function to estimate economy-wide energy efficiency performance from a production efficiency point of view. These types of Shephard distance functions are based on the concept of radial efficiency measure that assumes a proportional adjustment for all outputs or inputs. In literature related to DEA estimation, there exist a number of studies using DEA for energy efficiency analysis, e.g., [12–16]. For the shadow price or opportunity cost analysis of pollutants based on DEA, one may refer to [17,18].

Recently, a number of researchers have devoted to analyze the energy efficiency and the shadow prices of emissions in China by using distance function models. For instance, Hu and Wang [19] and Chang and Hu [20] use DEA to estimate regional energy efficiency for China. Wei et al. [21] analyze energy efficiency of China's iron and steel sector using DEA. Li [22] uses a distance function based on the DEA-Malmquist index to analyze carbon efficiency

change in Chinese provinces. Guo et al. [23] estimate the carbon efficiency and emission reduction potentials in different Chinese provinces based on an environmental DEA approach. Wang et al. [24] introduce a newly-developed efficiency measure considering CO₂ emissions in Chinese provinces based on linear programming. Zhang et al. [25] use the DEA-CCR model to estimate the industrial eco-efficiency of 30 Chinese provinces incorporating COD discharge, nitrogen, sulfur dioxide emission, soot emission, and dust emission as inputs. Shi et al. [26] use DEA-CCR/BCC models to measure Chinese regional industrial energy efficiency treating undesirable waste gases as inputs. Yeh et al. [27] conducted a comparative study on energy utilization efficiency between Taiwan and mainland China incorporating CO₂ and SO₂. Ke et al. [28] use the Shephard output distance function to estimate the shadow price of SO₂ of China. Kaneko et al. [29] employ directional distance function for SO₂ shadow price estimation of China. Ke et al. [30] use the directional output distance function to derive estimates of green inefficiency, shadow prices, and waste costs of three industrial wastes. As for the estimation of the CO₂ emission abatement cost in China, as far as we know, an example study is Wang et al. [31] in which the directional distance function and the non-parametric DEA approach are employed.

A common feature of previous studies for China is that they often adopt the radial efficiency measure. The radial measure allows for the equal and proportionate adjustment for all inputs or outputs. Despite its theoretical advantages, the radial measure has also several limitations. First, it does not provide information regarding the efficiency of the specific inputs or outputs included in the process [32]. Second, the radial measure has relatively weak discriminating power in ranking entities to be evaluated [33]. Third, the radial measure adjusts all undesirable outputs and inputs by the same proportion to the efficient targets. To overcome the weaknesses of the radial measure, some researchers have contributed to develop the non-radial efficiency measures. See, for example, [7,32–35]. Ref. [7] uses the parametric directional distance function method while [32–35] are based on the non-parametric DEA methods. Of the various non-radial DEA measures, the most typical and widely used one is referred to as the slack-based measure (SBM) proposed by Tone [34]. SBM directly deals with the “input excess” and “output shortfall,” and projects each entity to a ‘furthest’ point on the efficient frontiers in the sense that the objective function is to be minimized by finding the maximum slacks [34].

Following the idea of the SBM, Zhou et al. [18] develop a slacks-based DEA model in which undesirable outputs are incorporated to evaluate the impact of environmental regulation on economic efficiency. In this study, we extend [18] and employ an alternative slacks-based DEA model that incorporates specific slack adjustment for CO₂ emissions.⁴ Through the dual model of the slacks-based DEA model, the abatement costs of CO₂ emissions can be estimated.

3. Methodology

In this section, we first develop the potential CO₂ emissions reduction (PCR) index based on a SBM DEA model. In the second stage, the shadow prices of CO₂ emissions are derived from the dual model of the SBM DEA model.

3.1. CO₂ emissions efficiency and energy efficiency

In contrast to radial efficiency measures, which are based on the proportional reduction (enlargement) of input (output), SBM

² Hu et al. [5] use a life-cycle assessment to study the cost of CO₂ of automotive fuel in China.

³ Two kinds of distance function are widely used in literature. One is the Shephard distance function that is a radial efficiency measure and assumes a proportional adjustment for all outputs or inputs [6]. The other is directional distance function that is often regarded as a non-radial efficiency measure [7].

⁴ In energy studies, SBM-DEA has also been used to analyze energy efficiency incorporating undesirable outputs for China. For instance, Rao et al. [36] employed SBM-DEA to estimate China's energy efficiency incorporating pollutants COD and SO₂.

deals directly with input excess (potential reduction) and output shortfall (potential expansion) of the observation, called slack variables. The SBM projects the observations to the furthest point on the efficient frontier, in the sense that the objective function is to be minimized by finding the maximum amounts of slacks. Fig. 1 shows the difference between radial efficiency measure and SBM. We assume that the piecewise linear frontier is the combinations of energy and other inputs that produce the same total output. All points on the frontier are technically efficient. The point within the isoquant curve is inefficient because they use more inputs to produce the same output. Therefore, point “c” is not an efficient point. The radial efficiency measures adjust point “c” to efficient point “a” through origin and measures its efficiency score as “oa/oc.” A major problem with the radial efficiency method is that it does not provide efficiency of the specific inputs and all of the inputs’ efficiencies are measured at the same proportion “oa/oc.” For instance, the energy efficiency score is measured by “oe/of” equals to “oa/oc”. On the other hand, SBM is a non-radial efficiency measure that deals directly with the input excess and output shortfall of the observation, known as slack variables. It projects the observations to the “furthest point” on the efficient frontier, as shown in Fig. 1. Assuming that the furthest point in the frontier from point “c” is point “b”,⁵ in which case the energy slack (ES) is “df” referring to the potential reduction from the real energy input “of” to the target energy input “od”. Our SBM energy efficiency of point “c” is measured by “od/of.” Which is smaller than its radial score “oe/of”.

After graphically illustrating the SBM model, we will discuss it mathematically. Consider a production technology in which there are n Chinese province and each province employs capital (k), labor (l) and energy (e) to produce GDP (g) and CO₂ emissions (c) as a by-product of energy consumption. The production technology set can be defined as:

$$T = \{(k, l, e, g, c) : (k, l, e) \text{ can produce } (g, c)\} \quad (1)$$

In production economic theory T is often assumed to satisfy the standard axioms, e.g., inactivity is always possible and finite amounts of inputs can only produce finite amounts of outputs. Also, inputs and desirable outputs in T are assumed to be strongly or freely disposable. That is to say, if $(k, l, e, g, c) \in T$ and $(k', l', e') \geq (k, l, e)$ (or $g' \leq g$) then $(k', l', e', g, c) \in T$ (or $(k, l, e, g', c) \in T$). Once the production technology is specified, the parametric distance function or the non-parametric DEA method can be used to evaluate the environmental performance and shadow price of undesirable output. Due to the limitations of radial efficiency measures discussed above, the slack-based efficiency measure is considered to use in this paper. In a framework of joint production of desirable and undesirable outputs, Zhou et al. [18] develop a SBM model for measuring environmental performance. However, the model proposed in [18] cannot estimate the carbon efficiency from the target CO₂ emissions since it does not include the slack variable for CO₂ emissions. Therefore, instead of assuming the weak disposability of CO₂ emissions, we add the slack for CO₂ emissions into the objective function by modifying the constraint for CO₂ emissions. The following model can be derived:

$$\rho_0^* = \min \frac{1 - \frac{1}{3} \left(\frac{s_0^k}{k_0} + \frac{s_0^l}{l_0} + \frac{s_0^c}{e_0} \right)}{1 + \frac{1}{2} \left(\frac{s_0^g}{g_0} + \frac{s_0^c}{c_0} \right)}$$

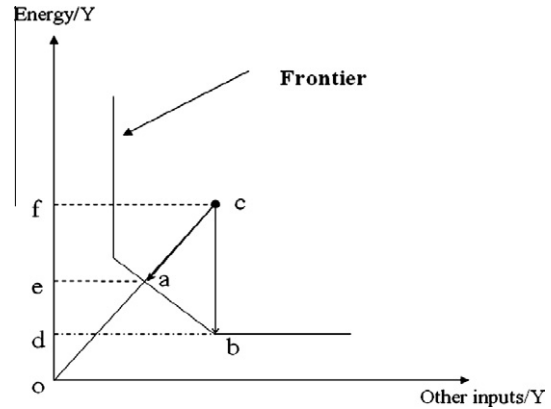


Fig. 1. Graphical illustration of radial efficiency measure vs. SBM-efficiency measure.

s.t.

$$k_0 = \sum_{j=1}^n \lambda_j k_j + s_0^k$$

$$l_0 = \sum_{j=1}^n \lambda_j l_j + s_0^l$$

$$e_0 = \sum_{j=1}^n \lambda_j e_j + s_0^e \quad (2)$$

$$g_0 = \sum_{j=1}^n \lambda_j g_j - s_0^g$$

$$c_0 = \sum_{j=1}^n \lambda_j c_j + s_0^c$$

$$s_0^k \geq 0, s_0^l \geq 0, s_0^e \geq 0, s_0^g \geq 0, s_0^c \geq 0, \lambda \geq 0$$

In Eq. (2), $\lambda = [\lambda_1 \dots \lambda_n] \in \mathbb{R}^n$ is a non-negative intensity vector. It should be pointed out that Eq. (2) is a constant return to scale (CRS) DEA model. One can impose some constraints on the intensity vector such as $\sum_{j=1}^n \lambda_j = 1$ for the variable returns to scale (VRS). Zhou and Ang [35] show that CRS assumption satisfies all production technologies and has more discrimination ability than that of the corresponding VRS model. “ s_0^g ” denotes the shortage of good outputs, and s_0^k, s_0^l, s_0^e and s_0^c refer to excesses of inputs k, l, e and undesirable output c . The subscript “0” means the observation whose efficiency is being estimated. The subscript “ j ” represents the observation j . The observation is efficient in the presence of CO₂ emissions if $\rho^* = 1$, indicating that all the slack variables are zeros. Since Eq. (2) is a nonlinear programming model, following the transformation suggested by [34], we can establish its equivalent linear programming model as follows:

$$r_0^* = \min t - \frac{1}{3} \left(\frac{s_0^k}{k_0} + \frac{s_0^l}{l_0} + \frac{s_0^c}{e_0} \right)$$

s.t.

$$1 = t + \frac{1}{2} \left(\frac{s_0^g}{g_0} + \frac{s_0^c}{c_0} \right)$$

$$tk_0 = \sum_{j=1}^n \eta_j k_j + s_0^k$$

$$tl_0 = \sum_{j=1}^n \eta_j l_j + s_0^l$$

$$te_0 = \sum_{j=1}^n \eta_j e_j + s_0^e$$

⁵ In the real case, the “furthest point” can be solved (found) by using a linear programming software packages.

$$\begin{aligned}
tg_0 &= \sum_{j=1}^n \eta_j g_j - S_0^g \\
tc_0 &= \sum_{j=1}^n \eta_j c_j + S_0^c \\
S_0^k &\geq 0, S_0^l \geq 0, S_0^e \geq 0, S_0^g \geq 0, S_0^c \geq 0, \eta \geq 0, t > 0.
\end{aligned} \quad (3)$$

In Eq. (3), the variables maintain the following relationship $\rho^* = r^*$, $\eta^* = \lambda^* * t^*$, $S_0^{k*} = S_0^k * t^*$, $S_0^{l*} = S_0^l * t^*$, $S_0^{e*} = S_0^e * t^*$, $S_0^{g*} = S_0^g * t^*$, $S_0^{c*} = S_0^c * t^*$ with model (2). Linear-transformed Model (3) can guarantee the optimal solution of $(t^*, \eta^*, S_0^{k*}, S_0^{l*}, S_0^{e*}, S_0^{g*}, S_0^{c*})$ and can be derived by employing the linear programming approach. If the observation is carbon-inefficient, its efficiency can be improved by reducing the excesses of S_0^c in carbon emission. Therefore, the CO₂ emissions reduction (PCR) of a given specific region can be measured as

$$PCR_0 = S_0^c = \frac{S_0^c}{t} \quad (4)$$

Meanwhile, the CO₂ emissions efficiency can be measured by the ratio of target CO₂ emissions to real CO₂ emissions:

$$CO_2 \text{ emissions efficiency} = \frac{c_0 - S_0^c}{c_0} \quad (5)$$

It should be pointed out that the “ratio of target to real” idea is introduced by Hu and Wang [19] in measuring energy efficiency, which has been further developed by Zhou and Ang [35] through incorporating undesirable outputs in energy efficiency analysis. Hence, the regional potential energy reduction (PER) and energy efficiency index can also be computed by

$$PER_0 = S_0^e = \frac{S_0^e}{t} \quad (6)$$

$$\text{Energy efficiency} = \frac{e_0 - S_0^e}{e_0} \quad (7)$$

3.2. Marginal abatement costs of CO₂ emissions

The shadow price of CO₂ emissions can be estimated by the dual model of Eq. (3). The dual program of the model (3) in the CRS consumption can be expressed as follows, with the dual variables $\sigma, p^k, p^l, p^e, p^g, p^c$.

$$\sigma^* = \max \sigma$$

s.t.

$$\begin{aligned}
\sigma + p^k k_0 + p^l l_0 + p^e e_0 - p^g g_0 + p^c c_0 &= 1 \\
-(\sum_{j=1}^n p^k k_j + \sum_{j=1}^n p^l l_j + \sum_{j=1}^n p^e e_j) + \sum_{j=1}^n p^g g_j - \sum_{j=1}^n p^c c_j &\leq 0 \\
p^k &\geq \frac{1}{3} [1/k_0], p^l \geq \frac{1}{3} [1/l_0], p^e \geq \frac{1}{3} [1/e_0] \\
p^g &\geq \frac{g}{2} [1/g_0], p^c \geq \frac{c}{2} [1/c_0]
\end{aligned} \quad (8)$$

In Eq. (8), p^k, p^l, p^e, p^g and p^c are dual variables correspond to the constraints “ $tk_0 = \sum_{j=1}^n \eta_j k_j + S_0^k, tl_0 = \sum_{j=1}^n \eta_j l_j + S_0^l, te_0 = \sum_{j=1}^n \eta_j e_j + S_0^e, tg_0 = \sum_{j=1}^n \eta_j g_j - S_0^g$ and $tc_0 = \sum_{j=1}^n \eta_j c_j + S_0^c$, respectively” in model (3) and σ is a dual variable correspond to the objective function of model (3).

According to [34], by the constraint “ $\sigma + p^k k_0 + p^l l_0 + p^e e_0 - p^g g_0 + p^c c_0 = 1$ ” in Eq. (8), the above problem is equivalent to the following:

$$\begin{aligned}
\max & p^g g_0 - (p^k k_0 + p^l l_0 + p^e e_0) - p^c c_0 \\
\text{s.t.} & \\
& \sum_{j=1}^n p^g g_j - \left(\sum_{j=1}^n p^k k_j + \sum_{j=1}^n p^l l_j + \sum_{j=1}^n p^e e_j \right) - \sum_{j=1}^n p^c c_j \leq 0 \\
& p^k \geq \frac{1}{3} [1/k_0], p^l \geq \frac{1}{3} [1/l_0], p^e \geq \frac{1}{3} [1/e_0] \\
& p^g \geq \frac{1 + p_0^g g_0 - (p^k k_0 + p^l l_0 + p^e e_0) - p_0^c c_0}{2} [1/g_0], \\
& p^c \geq \frac{1 + p_0^g g_0 - (p^k k_0 + p^l l_0 + p^e e_0) - p_0^c c_0}{2} [1/c_0]
\end{aligned} \quad (9)$$

The dual variable p^c can be interpreted as the shadow prices of CO₂ emissions (SPC), whereas p^g denotes the shadow price of desirable output GDP. Assuming that the absolute shadow price of GDP is equal to its market price (converted into 1 US \$ Constant 2000), the relative shadow price of CO₂ emissions (SPC) with regard to GDP can be measured by

$$SPC = 1\$ * \frac{p^c}{p^g} \quad (10)$$

The similar idea can be found in [17], which estimate a DEA model of electricity and SO₂ production and use the ratio of the dual values of the desirable and undesirable output constraints as the shadow price. In other word, the shadow price of CO₂ emissions can be interpreted as the marginal rate of transformation between CO₂ emissions and GDP. DMUs are not able to freely abate CO₂ emissions. That is, they incur the opportunity cost of reducing GDP resulting from the diversion of some inputs for emission abatement efforts. In fact, cleaning up CO₂ emissions actually requires less fossil fuel consumption. DMUs invest either in improving boiler fuel-efficiency or in increasing dependence on renewable sources [37]. Model (9) aims at maximizing the virtual profit, “ $p^g g_0 - (p^k k_0 + p^l l_0 + p^e e_0) - p^c c_0$ ” for the observation concerned. This is a typical profit maximization model in which the virtual profit is at best zero (will not exceed 0) when $\sigma = 1$ for the slack-based measure efficient observation. The Lp_solve 5.5 software package is used to solve the linear programming of our models.

4. Empirical study

4.1. Data

The models described in Section 3 have been applied to study the efficiency, reduction potentials and marginal abate costs for CO₂ emissions in different provinces of China during 2001–2010. We first collect the data for the inputs and outputs described in our framework. Since the data for Tibet cannot be obtained, our dataset covers thirty provinces. The data on GDP and labor can be found in [38]. Capital stock data is from [39] and the data after 2006 is estimated by perpetual inventory method as given in Eq. (11) suggested by [19–20]. All the monetary variables including GDP and capital stock have been converted into 2000 constant prices with GDP deflators and then transformed into US\$ based on exchange rate.⁶

$$K_t = K_{t-1} + KF - KD \quad (11)$$

Note: K_t is the Capital stock (current year); K_{t-1} is the Capital stock (previous year); KF is the Capital formation (current year); KD is the Capital depreciation (current year).

⁶ Chinese currency (yuan) is converted into US \$ value by the 2000 exchange rate (RMB 8.278 yuan to one dollar).

Energy consumption is selected as energy input, including all types of energy such as coal, oil, and gas. All of them have been converted into tons of standard coal equivalent (TCE) in terms of the corresponding energy folding standard.⁷ The official data on provincial CO₂ emissions is not available in China. Following [22–24], we employ the fuel-based carbon calculation model described [40,41] to estimate the provincial CO₂ emissions. The amounts of energy consumption by fuel type can be found in [42].

Table 1 shows the descriptive statistics of the regional data. The mean value of capital input (*k*) was 93 (Billion \$ US dollars based on constant year 2000 price), and that of labor input (*l*) was 5659 (thousands of persons). The fossil energy consumption input (*e*) was 84,698 (1000 tons of coal equivalent or TCE). Desirable output GDP (*g*) was 77 (Billion \$ US dollars based on constant year 2000 price), whereas the undesirable outputs of CO₂ (*c*) were 177,866 (thousands of Tons).

Table 2 shows the correlation matrix of outputs and inputs. It clearly shows that the correlation coefficients between inputs and outputs are all significantly positive. Thus, when inputs are increased, the outputs tend to increase as well. This formulation is, then, suitable for statistical efficiency analysis.

4.2. CO₂ emission efficiency and energy efficiency

According to the classifications of the Chinese regional distribution, China's provinces are divided into three regions, namely the east, central and west areas. The east area is composed of eleven regions: eight coastal provinces (e.g., Shandong, Jiangsu, Zhejiang, and Guangdong) and three municipalities (Beijing, Tianjin, and Shanghai). This area enjoys great progress in economy development and its GDP output is around half of Chinese total GDP. The central area was composed of ten inland provinces (e.g., Heilongjiang, Jilin, and Inner Mongolia). This area has a large population and it is a home base for farming and related industries. The economy growth is less than east area. The west area covers more than half of China's territory and includes one municipality (Chongqing) and nine provinces (e.g., Gansu, Qinghai, Xinjiang, and Sichuan). Compared with the other two areas, this area is the least developed area in China, and has a lower population density. Using Eqs. (4) and (5), we can estimate the CO₂ emission potential reduction and the CO₂ emission efficiency, respectively, as reported in Tables 3 and 4.

With regard to the potential CO₂ emission reduction (PCR), the average PCR can be calculated as 56.1 million tons for each province and 1683 million tons in total. In 2010, Shanghai, Guangdong, Fujian and Jiangsu evidence no potential reduction, indicating that these four provinces exhibit the best CO₂ emission efficiency values. Hebei shows the largest potential for CO₂ emission reduction at 305.5 million tons. Shandong, which ranks second in terms of PCR, also evidences great potential CO₂ emission reduction (276.6 million tons). Hebei and Shandong are the two largest energy consumption provinces, and have profound carbon emission reduction potential.

With regard to the CO₂ emission efficiency, the results demonstrate that most of the provinces in China are not performing CO₂-efficiently, as they employ massive carbon-based energy resources in production process. For instance, in 2010, CO₂ emission efficiency varies from 0.146 to 1, with an average value of 0.645, indicating that the 30 provinces, on average, could accomplish approximately a 35.5% CO₂ emission reduction if the province operates on the frontier of production technology. This, in turn, leads to the conclusion that China's regional economy is not, in fact, a low-carbon regime. Thus, most provinces appear to have the potential to reduce amounts of carbon emissions.

Table 1

Descriptive statistics of inputs and outputs, 2001–2010. (N = 300).

Inputs & outputs	Variable	Unit	Mean	Max	Min	Std. dev.
Non-energy inputs	Capital	10 ⁹ US\$ ^a	93	466	2	84
	Labor	10 ³ persons	5659	56,624	148	7643
Energy inputs	Energy	10 ³ TCE	84,698	389,447	4799	63,569
	Desirable output	10 ⁹ US\$ ^a	77	381	2	70
Undesirable	CO ₂	10 ³ tons	177,866	817,839	10,078	133,495

^a Capital and GDP are converted into US\$ based on 2000 exchange rate (1\$ = 8.278 Y).

Table 2

Correlation matrixes for inputs and outputs.

	Capital	Labor	Energy	GDP	CO ₂
Capital	1				
Labor	0.218 ^a	1			
Energy	0.887 ^a	0.325 ^a	1		
GDP	0.945 ^a	0.303 ^a	0.912 ^a	1	
CO ₂	0.898 ^a	0.314 ^a	0.974 ^a	0.903 ^a	1

^a Represents the significance at 5% level.

It is intuitively interesting that the east area evidences the highest CO₂ emission efficiency score of 0.841, followed by the central area with an efficiency score of 0.625. The west area shows the lowest efficiency score of 0.445. This regional difference is shown to be statistically significant by the non-parametric Kruskal–Wallis test, and the Chi-square value is 16.38 at a significance level of 1%. The results indicate that the east area evidences better low-carbon-economic performance than the other two regions, and the economically highly developed east area also evidences higher CO₂ emission efficiency. From the comparative regional characteristics of economic development, this regional difference is very important in that the economically well-developed regions evidence better CO₂ emission efficiency, and thus this type of economic dichotomy may result in serious problems in nationwide equitable development [44]. In order to overcome this regional discrepancy, the equitable market mechanism of CO₂ emission abatement cost may be a desirable solution. Therefore, the appropriate feasibility of CO₂ emission abatement cost should be handled with great caution.

Using Eqs. (6) and (7), we can also compute the energy potential reduction and the energy efficiency, respectively, as reported in Tables 5 and 6.

With regard to the potential energy reduction (PER) based on slack variable of energy, the average PER can be computed as 35.8 million tons for each province and 1074 million tons in total. For instance, in 2010, Shanghai, Guangdong, and Jiangsu evidence no potential reduction, indicating that these three provinces exhibit the best energy efficiency values. Hebei shows the largest potential for energy at 132.3 million tons. Shandong, which ranks second in terms of PER, also evidences great energy reduction (127.8 million tons). Hebei and Shandong are the two largest energy consumption provinces, and have profound energy use reduction potential.

With regard to the energy efficiency reported in Table 6, the results indicate that most provinces are not performing energy-efficiently, as they employ massive fossil energy resources in production process. For instance, in 2010, energy efficiency varies from 0.202 to 1, with an average value of 0.575 a little lowers than CO₂ emission efficiency, indicating that the 30 provinces, on average, could accomplish approximately a 42.5% energy reduction if the province operates on the frontier of production technology.

⁷ The residential energy consumption is excluded from total energy input because this part does not come from production process energy consumption [43].

Table 3Potential reduction of CO₂ emissions for 30 provinces, 2001–2010.

Provinces	2001	2002	2003	2004	2005	2006	2007	2008	2009	2010
Beijing	26.5	24.7	24.1	24.3	22.6	17.1	0.0	2.5	0.0	9.7
Tianjin	20.4	17.8	16.1	18.1	16.5	16.8	22.3	16.6	12.0	28.3
Hebei	97.6	91.2	141.3	160.9	222.7	197.6	204.9	195.8	211.7	305.5
Shanxi	87.3	87.2	136.3	146.7	172.7	183.6	195.3	191.8	185.7	260.7
Inner Mongolia	36.0	36.4	62.7	86.9	117.8	124.2	137.6	145.6	150.3	246.0
Liaoning	106.4	104.0	102.9	101.6	130.5	100.3	103.3	99.2	113.3	197.5
Jilin	34.6	34.7	54.3	56.9	58.8	47.1	50.3	50.7	55.5	86.4
Heilongjiang	29.8	0.0	32.6	32.6	68.5	59.7	51.3	50.3	42.9	52.2
Shanghai	18.9	13.0	20.9	16.8	21.2	14.1	1.7	0.0	0.0	0.0
Jiangsu	1.7	0.0	2.7	9.1	110.3	34.5	41.5	0.0	0.0	0.0
Zhejiang	14.3	3.3	24.7	26.6	74.2	43.5	31.9	16.9	0.2	50.8
Anhui	28.3	26.2	32.6	29.0	49.7	32.5	24.3	30.5	24.6	63.0
Fujian	0.0	0.0	0.0	0.0	20.9	0.0	0.0	0.0	0.0	0.0
Jiangxi	11.8	9.5	14.0	14.7	27.6	18.4	16.8	12.5	7.0	34.0
Shandong	72.0	44.0	102.2	126.9	244.1	174.4	179.2	157.7	178.5	276.6
Henan	55.4	28.2	62.6	95.5	132.2	98.6	98.8	91.9	78.1	153.0
Hubei	23.5	18.6	40.4	48.8	82.0	52.0	50.0	39.2	43.2	104.5
Hunan	7.8	9.2	22.8	40.9	91.5	67.6	68.1	60.5	55.1	129.8
Guangdong	0.0	0.0	0.0	0.0	83.3	0.0	0.0	0.0	0.0	0.0
Guangxi	20.8	13.3	22.2	30.3	47.7	38.6	40.2	34.5	43.2	75.3
Hainan	0.0	1.7	0.5	0.2	0.3	2.4	0.3	0.0	0.0	3.3
Chongqing	0.0	10.1	23.7	27.3	52.7	49.4	48.4	48.8	48.0	70.0
Sichuan	34.9	29.8	63.9	77.9	105.9	79.2	79.3	76.6	69.3	136.2
Guizhou	55.0	57.6	74.2	80.1	73.5	76.1	82.6	80.9	84.4	127.5
Yunnan	25.5	26.0	36.9	45.9	66.2	60.9	58.8	57.9	57.7	90.3
Shaanxi	52.4	56.1	42.4	49.2	68.5	53.5	68.1	65.4	66.6	120.9
Gansu	18.5	18.8	30.3	33.4	40.7	39.7	41.2	41.0	36.2	60.5
Qinghai	0.0	0.0	11.4	14.6	20.2	21.5	23.4	24.9	24.7	40.9
Ningxia	17.7	21.0	32.9	38.2	42.6	45.8	49.4	51.0	52.6	94.2
Xinjiang	33.6	35.1	43.8	53.5	67.6	66.0	67.3	70.4	74.0	94.6

Note: Unit: million tons of CO₂.**Table 4**CO₂ emission efficiency 30 provinces, 2001–2010.

Province	Area	2001	2002	2003	2004	2005	2006	2007	2008	2009	2010
Shanxi	Central	0.256	0.277	0.247	0.252	0.223	0.253	0.282	0.298	0.316	0.273
Jilin	Central	0.540	0.561	0.474	0.491	0.446	0.601	0.616	0.648	0.638	0.563
Heilongjiang	Central	0.761	1.000	0.759	0.783	0.578	0.661	0.729	0.750	0.797	0.784
Anhui	Central	0.703	0.736	0.694	0.753	0.609	0.764	0.839	0.812	0.858	0.731
Jiangxi	Central	0.754	0.816	0.786	0.799	0.664	0.793	0.826	0.878	0.937	0.777
Henan	Central	0.621	0.812	0.680	0.605	0.511	0.671	0.700	0.738	0.786	0.678
Hubei	Central	0.802	0.848	0.723	0.717	0.570	0.751	0.782	0.839	0.833	0.707
Hunan	Central	0.899	0.887	0.811	0.718	0.507	0.666	0.694	0.744	0.784	0.638
Shaanxi	Central	0.048	0.051	0.496	0.489	0.390	0.567	0.501	0.563	0.589	0.475
Beijing	East	0.679	0.715	0.740	0.763	0.794	0.855	1.000	0.980	1.000	0.941
Tianjin	East	0.630	0.698	0.746	0.751	0.796	0.811	0.771	0.843	0.897	0.812
Hebei	East	0.517	0.557	0.489	0.487	0.378	0.498	0.519	0.554	0.539	0.476
Liaoning	East	0.466	0.487	0.511	0.584	0.487	0.642	0.666	0.702	0.683	0.585
Shanghai	East	0.827	0.885	0.846	0.886	0.870	0.920	0.991	1.000	1.000	1.000
Jiangsu	East	0.990	1.000	0.988	0.968	0.691	0.913	0.905	1.000	1.000	1.000
Zhejiang	East	0.894	0.976	0.874	0.880	0.700	0.840	0.893	0.946	0.999	0.876
Fujian	East	1.000	1.000	1.000	1.000	0.831	1.000	1.000	1.000	1.000	1.000
Shandong	East	0.689	0.813	0.698	0.683	0.504	0.680	0.699	0.747	0.730	0.662
Guangdong	East	1.000	1.000	1.000	1.000	0.774	1.000	1.000	1.000	1.000	1.000
Hainan	East	1.000	0.880	0.967	0.986	0.981	0.876	0.986	1.000	1.000	0.905
Inner Mongolia	West	0.418	0.443	0.377	0.346	0.301	0.365	0.382	0.408	0.438	0.369
Guangxi	West	0.625	0.777	0.697	0.653	0.528	0.655	0.678	0.745	0.707	0.635
Chongqing	West	1.000	0.805	0.608	0.624	0.460	0.533	0.588	0.618	0.654	0.603
Sichuan	West	0.703	0.755	0.616	0.597	0.504	0.662	0.691	0.720	0.765	0.656
Guizhou	West	0.280	0.279	0.249	0.255	0.270	0.309	0.319	0.360	0.375	0.310
Yunnan	West	0.607	0.621	0.556	0.529	0.412	0.507	0.559	0.587	0.616	0.535
Gansu	West	0.606	0.623	0.447	0.450	0.401	0.462	0.481	0.507	0.575	0.475
Qinghai	West	1.000	1.000	0.393	0.363	0.281	0.328	0.335	0.350	0.374	0.294
Ningxia	West	0.240	0.230	0.172	0.166	0.150	0.179	0.186	0.200	0.213	0.146
Xinjiang	West	0.453	0.462	0.433	0.410	0.335	0.410	0.446	0.461	0.468	0.432
–	Central mean	0.598	0.665	0.630	0.623	0.500	0.636	0.663	0.697	0.727	0.625
–	East mean	0.790	0.819	0.805	0.817	0.710	0.821	0.857	0.888	0.895	0.841
–	West mean	0.593	0.600	0.455	0.439	0.364	0.441	0.467	0.495	0.518	0.445
–	China mean	0.667	0.700	0.636	0.633	0.532	0.639	0.669	0.700	0.719	0.645

Table 5
Potential reduction of energy use for 30 provinces, 2001–2010.

Provinces	2001	2002	2003	2004	2005	2006	2007	2008	2009	2010
Beijing	24.1	24.7	25.7	27.3	31.2	20.6	0.0	18.5	14.6	20.9
Tianjin	14.6	14.6	14.6	15.5	20.2	15.4	18.2	13.4	13.3	14.1
Hebei	55.9	54.2	75.1	83.1	119.0	99.1	92.7	101.4	106.7	132.3
Shanxi	38.8	39.5	61.1	66.1	82.9	84.3	86.5	94.4	96.1	125.6
Inner Mongolia	17.9	18.6	31.0	41.6	60.4	61.1	66.0	72.9	76.8	110.6
Liaoning	41.7	39.3	41.3	53.5	73.0	61.0	58.3	68.1	71.4	93.5
Jilin	20.8	21.6	29.4	31.2	33.5	29.0	33.9	35.8	37.4	47.1
Heilongjiang	14.1	0.0	13.7	11.7	39.7	25.3	36.0	35.1	37.0	37.2
Shanghai	22.4	20.1	23.1	20.5	35.5	23.0	12.7	0.0	0.0	0.0
Jiangsu	25.3	18.0	27.4	34.2	91.5	48.5	40.7	21.2	5.0	0.0
Zhejiang	20.0	16.5	31.4	36.6	68.1	48.8	42.8	39.7	39.5	52.8
Anhui	22.2	22.0	23.2	24.5	37.3	29.5	29.1	29.8	32.3	44.8
Fujian	3.4	0.0	8.2	9.5	25.6	9.7	7.8	3.1	0.0	3.7
Jiangxi	9.0	9.8	14.1	15.9	24.5	19.5	20.7	20.7	22.0	32.6
Shandong	39.7	36.5	67.7	81.5	146.5	80.4	114.7	107.2	109.2	127.8
Henan	32.5	31.7	45.4	58.1	83.2	70.4	76.7	79.1	81.1	107.4
Hubei	26.2	25.3	31.7	38.6	54.6	43.5	40.8	37.2	37.2	52.7
Hunan	10.6	11.6	23.5	29.5	56.2	42.5	47.0	46.7	50.5	74.1
Guangdong	0.0	0.0	0.0	0.0	71.4	0.0	0.0	0.0	0.0	0.0
Guangxi	13.8	14.2	18.3	22.4	32.8	27.6	30.3	31.8	34.6	51.0
Hainan	1.7	3.2	2.8	2.8	3.9	4.7	2.9	1.3	0.0	2.5
Chongqing	0.0	6.3	13.7	18.2	33.0	30.2	32.7	35.1	37.6	46.7
Sichuan	28.9	28.5	42.8	49.6	69.1	56.0	61.1	63.5	66.7	88.6
Guizhou	26.5	27.9	35.2	38.4	39.3	38.6	42.3	43.4	46.6	63.9
Yunnan	18.5	19.2	24.2	28.3	40.2	37.3	39.4	40.8	43.3	56.4
Shaanxi	20.4	21.9	26.4	30.4	40.4	20.6	43.2	46.2	49.8	72.2
Gansu	12.8	13.6	17.6	19.4	24.5	23.5	24.8	25.7	26.1	36.1
Qinghai	0.0	0.0	6.4	7.8	10.9	11.1	12.0	12.8	13.3	19.6
Ningxia	8.5	10.0	15.3	17.7	20.8	21.6	23.3	24.2	25.3	41.9
Xinjiang	20.4	21.5	25.7	30.6	37.2	37.3	39.9	42.3	45.3	53.8

Note: (Unit: million TCE).

Table 6
Energy efficiency 30 provinces, 2001–2010.

Provinces	Area	2001	2002	2003	2004	2005	2006	2007	2008	2009	2010
Shanxi	Central	0.305	0.313	0.291	0.293	0.217	0.279	0.332	0.275	0.257	0.264
Jilin	Central	0.420	0.427	0.402	0.415	0.336	0.484	0.456	0.478	0.488	0.500
Heilongjiang	Central	0.762	1.000	0.787	0.837	0.487	0.698	0.600	0.634	0.632	0.677
Anhui	Central	0.512	0.534	0.543	0.562	0.383	0.552	0.596	0.615	0.609	0.598
Jiangxi	Central	0.603	0.603	0.549	0.542	0.372	0.541	0.549	0.576	0.584	0.552
Henan	Central	0.534	0.557	0.513	0.495	0.353	0.507	0.511	0.526	0.533	0.526
Hubei	Central	0.536	0.566	0.543	0.530	0.398	0.563	0.626	0.678	0.698	0.690
Hunan	Central	0.715	0.703	0.591	0.573	0.364	0.559	0.556	0.585	0.583	0.566
Shaanxi	Central	0.222	0.223	0.341	0.337	0.245	0.649	0.335	0.352	0.355	0.342
Beijing	East	0.387	0.400	0.418	0.440	0.405	0.632	1.000	0.692	0.766	0.732
Tianjin	East	0.445	0.481	0.517	0.554	0.474	0.636	0.608	0.734	0.759	0.803
Hebei	East	0.420	0.447	0.429	0.443	0.302	0.471	0.543	0.515	0.512	0.523
Liaoning	East	0.560	0.593	0.588	0.541	0.397	0.543	0.604	0.570	0.580	0.588
Shanghai	East	0.572	0.630	0.642	0.709	0.546	0.728	0.862	1.000	1.000	1.000
Jiangsu	East	0.704	0.794	0.750	0.747	0.461	0.743	0.804	0.904	0.979	1.000
Zhejiang	East	0.689	0.751	0.664	0.655	0.423	0.623	0.700	0.732	0.741	0.729
Fujian	East	0.898	1.000	0.822	0.818	0.566	0.852	0.893	0.961	1.000	0.967
Shandong	East	0.640	0.675	0.580	0.572	0.375	0.690	0.595	0.638	0.653	0.672
Guangdong	East	1.000	1.000	1.000	1.000	0.593	1.000	1.000	1.000	1.000	1.000
Hainan	East	0.649	0.528	0.589	0.623	0.520	0.491	0.728	0.887	1.000	0.847
Inner Mongolia	West	0.391	0.401	0.354	0.342	0.247	0.344	0.377	0.377	0.397	0.404
Guangxi	West	0.478	0.501	0.475	0.463	0.320	0.484	0.489	0.505	0.506	0.480
Chongqing	West	1.000	0.744	0.526	0.472	0.290	0.401	0.414	0.424	0.432	0.444
Sichuan	West	0.484	0.507	0.459	0.461	0.320	0.499	0.501	0.513	0.525	0.530
Guizhou	West	0.272	0.266	0.252	0.250	0.180	0.265	0.268	0.279	0.275	0.273
Yunnan	West	0.399	0.411	0.389	0.391	0.250	0.367	0.379	0.390	0.394	0.390
Gansu	West	0.424	0.427	0.327	0.328	0.242	0.329	0.343	0.351	0.356	0.342
Qinghai	West	1.000	1.000	0.287	0.282	0.185	0.271	0.284	0.295	0.290	0.289
Ningxia	West	0.234	0.232	0.192	0.187	0.128	0.188	0.195	0.204	0.204	0.202
Xinjiang	West	0.303	0.306	0.300	0.293	0.232	0.300	0.310	0.320	0.316	0.322
Central Mean		0.512	0.547	0.506	0.509	0.351	0.537	0.507	0.524	0.527	0.524
East Mean		0.633	0.664	0.636	0.646	0.460	0.674	0.758	0.785	0.817	0.806
West Mean		0.499	0.480	0.356	0.347	0.239	0.345	0.356	0.366	0.369	0.368

After estimating the CO₂ emission efficiency and energy efficiency, the regions can be categorized into four groups by high/low energy and CO₂ emission efficiency as shown in Table 7, the efficiency scores are based on the average value of 2001–2010, the top 15 provinces belong to the high efficiency group. The table indicates that the high/low CO₂ emission efficiency groups share the same members with the high/low energy efficiency group except for Henan and Liaoning. Henan belongs to high CO₂ emission efficiency but low energy efficiency group, Liaoning belongs to low CO₂ efficiency group but high energy efficiency group.

The results also indicate a regional difference of energy efficiency. The east area evidences the highest energy efficiency score of 0.806, followed by the central area with an efficiency score of 0.524. The west area shows the lowest efficiency score of 0.368. This regional difference is also statistically significant by the non-parametric Kruskal–Wallis test, and the Chi-square value is 15.29 at a significance level of 1%. The results indicate that the east area evidences both better energy efficiency performance and higher CO₂ emission efficiency than the other two regions. The results are similar from some previous studies such as [19,23–26,45,46]. Refs. [19,26] showed that the east area has a higher energy efficiency than other two areas, and Refs. [23–25,45,46] found that the east area has a higher environmental efficiency incorporating three industrial wastes. The results indicate that the east area enjoys both high economic achievement and high energy efficiency/CO₂ emission efficiency. This can be interpreted as follows: the east area is a more developed and economically advanced area, thus, the government can allocate more of the capital derived from the area's rapid economic growth to the area's environmental governance and energy efficiency technology for its sustainable development.

Regarding CO₂ emission efficiency and the energy efficiency trend of China from 2001 to 2010, Fig. 2 shows that the both average CO₂ emission efficiency and energy efficiency was decreasing during 2001–2005; however, from 2006, the efficiencies showed a continual increase. The “U-shape” trend may be interpreted as follows:

Before 2006, China's economic growth was fueled by energy-intensive heavy industry and infrastructure building, which required enormous amounts of energy. Moreover, the government did not pay much attention to energy intensity. Since 2006, China's 11th Five-Year Economic Plan (2006–2010) clearly illustrates the importance of building an energy-efficient and environmentally friendly society, and the Chinese government announced the goal of a 20% reduction in energy intensity (energy consumption per GDP) between 2006 and 2010. Therefore, it is likely that the

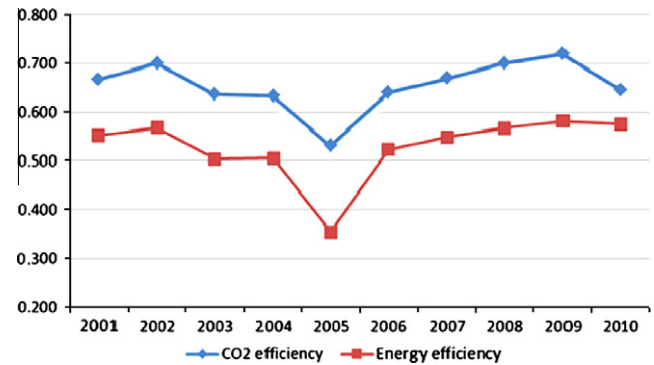


Fig. 2. The trend of CO₂ efficiency and energy efficiency.

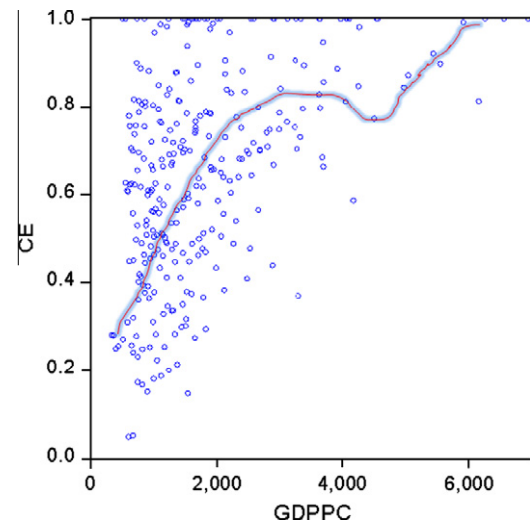


Fig. 3. The relationship between GDP per capita and CO₂ emission efficiency.

government's energy policy increased the CO₂ emission and energy efficiency.

Comparing with CO₂ emission efficiency and energy efficiency, we can find that, the energy efficiency is lower than the CO₂ emission efficiency indicating that improving the energy efficiency is much urgent for China.

Table 7
Groups by high/low energy and CO₂ emission efficiency.

	High CO ₂ efficiency group	High Energy efficiency group	Low CO ₂ efficiency group	Low Energy efficiency group
Provinces	Henan (15) Shandong(14) Hunan(13) Anhui(12) Hubei(11) Heilongjiang(10) Tianjin (9) Jiangxi(8) Beijing(7) Zhejiang(6) Shanghai(5) Jiangsu(4) Hainan(3) Guangdong (2) Fujian (1)	Jiangxi(15) Anhui(14) Liaoning (13) Hunan(12) Hubei(11) Beijing(10) Tianjin(9) Shandong(8) Zhejiang(7) Hainan(6) Heilongjiang(5) Shanghai(4) Jiangsu(3) Fujian(2) Guangdong(1)	Ningxia(30) Shanxi(29) Guizhou(28) Innermongolia(27) Shaanxi(26) Xinjiang(25) Qinghai(24) Hebei(23) Gansu(22) Yunnan(21) Jilin(20) Liaoning (19) Chongqing(18) Sichuan(17) Guangxi(16)	Ningxia(30) Guizhou(29) Shanxi(28) Xinjiang(27) Shaanxi(26) Gansu(25) InnerMongolia(24) Yunnan(23) Qinghai(22) Jilin(21) Hebei(20) Guangxi(19) Sichuan(18) Henan (17) Chongqing(16)

Note: the number in the parenthesis is the efficiency ranking.

Table 8
Parameter estimation of environmental Kuznets curve.

Variable	OLS	Tobit	Truncated	Bootstrapped truncated
GDPPC	0.000346*** (5.52517)	0.000353*** (4.83502)	0.000311*** (4.14205)	0.000311*** (4.34125)
GDPPC ²	−8.20E−08*** (−3.52028)	−8.63E−08*** (−3.03422)	−6.86E−08** (−2.34157)	−6.86E−08*** (−2.70472)
GDPPC ³	6.51E−12** (2.55326)	7.61E−12** (2.32219)	5.51E−12* (1.65134)	5.51E−12** (2.04231)
Servs	0.007698*** (6.17968)	0.007812*** (5.53102)	0.008418*** (5.85426)	.008418*** (5.31468)

The values within the parentheses refer to *t*-values.

2000 Replicates are used for bootstrapped truncated model.

* Mean the significance level 10%.

** Mean the significance level 5%.

*** Mean the significance level 1%.

The environmental Kuznets curve hypothesis claims that there exists an inverted U-type relationship between the level of pollutants and income. In order to investigate the existence of a Kuznets curve relationship between CO₂ emission efficiency and income in China, we develop the following equation:

$$CE_i = \beta_1 + \beta_2 GDPPC_i + \beta_3 GDPPC_i^2 + \beta_4 GDPPC_i^3 + \beta_5 servs_i + \varepsilon_i \quad (12)$$

where *CE* is CO₂ emission efficiency, *i* is province index, ε_i is the disturbance term, *GDPPC* is GDP per capita, and *Servs* is share of service industry in GDP. The shape of the polynomial equation will expose the relationship between CO₂ emission efficiency and GDP per capita. A positive sign for *GDPPC* coupled with a negative sign for its quadratic and a positive sign for its cubic terms will imply an improving CO₂ emission efficiency at the initial phases of growth which is followed by a phase of deterioration and then a further

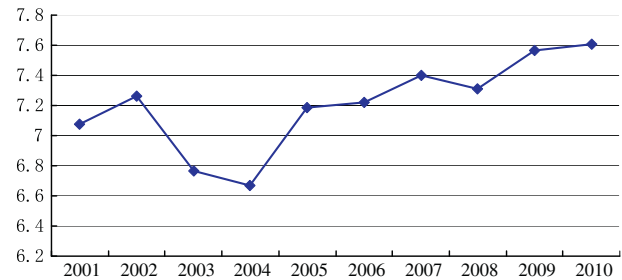


Fig. 4. Marginal abatement cost (MAC) of CO₂, 2001–2010.

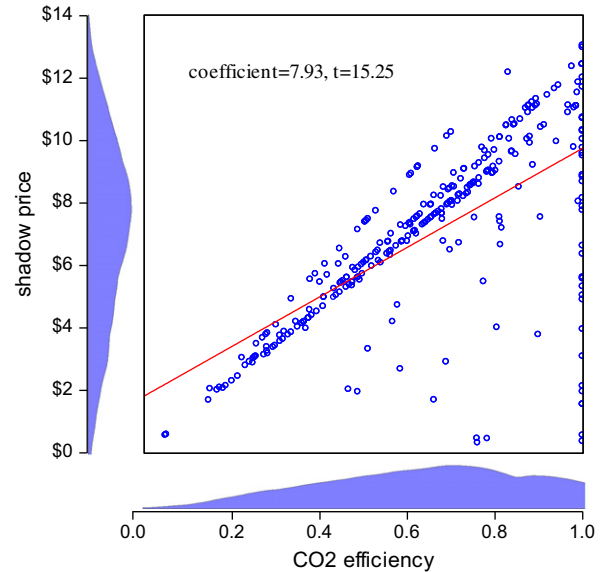


Fig. 5. The relationship between CO₂ shadow price and CO₂ emission efficiency.

Table 9
Marginal abatement cost of CO₂ emissions, 2001–2010 (unit: US \$).

Provinces	2001	2002	2003	2004	2005	2006	2007	2008	2009	2010
Beijing	7.65	8.06	8.33	8.60	8.95	8.51	10.05	9.79	5.03	9.96
Tianjin	7.64	8.02	8.57	8.64	8.96	9.32	9.77	10.49	11.16	10.10
Hebei	6.28	6.76	5.62	5.59	5.55	5.72	5.97	6.37	6.71	5.92
Shanxi	3.49	3.78	2.87	3.05	3.03	3.07	3.17	3.43	3.63	3.13
Inner Mongolia	5.69	6.04	4.58	4.19	4.09	4.20	4.39	4.69	5.03	3.97
Liaoning	2.03	1.95	3.32	2.67	7.14	7.38	7.65	8.07	8.50	7.28
Jilin	6.08	6.32	5.34	5.53	6.54	6.76	7.08	7.45	7.94	7.01
Heilongjiang	0.31	0.55	0.45	0.45	4.72	1.69	8.21	8.62	9.16	9.01
Shanghai	10.48	11.22	10.52	11.03	11.03	11.45	8.04	3.96	8.84	12.44
Jiangsu	11.86	10.70	11.53	11.13	10.13	10.49	10.43	2.13	6.36	5.32
Zhejiang	11.33	12.37	10.05	10.12	10.27	10.65	11.11	11.77	12.43	13.90
Anhui	7.92	8.29	8.42	8.65	8.93	9.27	9.65	10.11	10.68	9.09
Fujian	13.04	12.98	10.74	10.30	12.18	9.76	9.59	9.51	5.61	8.16
Jiangxi	7.53	7.19	9.53	9.69	9.73	10.05	10.47	11.13	11.66	9.67
Shandong	2.91	6.66	6.49	6.77	7.39	7.66	8.03	8.58	9.08	8.24
Henan	7.01	7.40	7.48	7.33	7.49	7.72	8.05	8.48	9.04	7.80
Hubei	9.03	9.55	8.77	8.24	8.35	8.63	8.99	9.64	10.05	8.52
Hunan	3.78	9.21	7.54	6.72	7.43	7.65	7.97	8.55	9.01	9.33
Guangdong	4.37	6.63	1.55	1.55	5.48	8.89	1.95	5.57	7.76	10.33
Guangxi	9.17	9.43	8.46	7.93	7.75	7.95	8.22	8.56	8.96	9.30
Hainan	12.02	9.91	10.89	11.11	11.05	11.11	11.10	7.89	3.09	7.54
Chongqing	4.90	4.01	7.30	7.57	6.27	6.47	6.75	7.11	7.52	7.93
Sichuan	8.53	9.16	7.47	7.24	7.38	7.61	7.94	8.27	8.79	7.54
Guizhou	3.25	3.38	3.02	3.09	3.68	3.75	3.87	4.14	4.31	3.56
Yunnan	7.37	7.54	6.75	6.41	6.05	6.16	6.43	6.75	7.08	7.15
Shaanxi	5.55	0.58	6.01	5.93	5.72	4.20	6.08	6.47	6.78	5.46
Gansu	8.88	9.14	5.42	5.46	5.46	5.60	5.84	6.15	6.61	5.46
Qinghai	11.71	8.68	4.52	4.18	3.83	3.77	3.85	4.02	4.30	3.38
Ningxia	6.91	6.79	6.08	6.01	6.04	6.06	6.14	6.29	6.45	6.68
Xinjiang	5.50	5.61	5.25	4.97	4.92	4.97	5.13	5.30	5.38	4.97
Mean	6.94	7.13	6.63	6.54	7.05	7.08	7.26	7.28	7.43	7.44

improvement once a critical level of per capita GDP is reached as Zaim and Taskin [47] suggested. Fig. 3 illustrates this relationship between CO₂ emission efficiency and GDP per capita. As the dependent variable *CE* is left-censored at zero and right censored at one, the OLS estimation may give biased estimation so that the Tobit regression is often employed. Moreover, according to Simar and Wilson [48], the truncated regression and bootstrapped truncated model has relatively unbiased estimation compared to Tobit regression. In this study, all of the four estimation methods have been used. The estimation results from the four methods are shown in Table 8. It is found that in the four models all the coefficients for *GDPPC* are shown statistical significance confirming the above environmental Kuznets curve hypothesis. Variable *Servs* is significant indicating that the share of service industry in GDP has a positive impact on CO₂ emission efficiency.

4.3. CO₂ emission abatement cost

As reported in Table 9, the estimated shadow prices for CO₂ emissions calculated via (7) and (8) can be interpreted as measures of opportunity cost of CO₂ emission abatement in terms of GDP converted into dollar values per unit of CO₂ emissions abatement. Therefore, the shadow prices measure the marginal abatement cost of CO₂ emission to the regional economy and to society in general. Table 8 demonstrates that the 30 provinces could pay, on average, \$7.2 to abate one ton of CO₂ emissions during 2001–2010.

The CO₂ emission shadow price of China in this study is lower than the current international market price of CO₂ emissions trading. The current market price of carbon emission trading is 12.22 €/T in the EU market according to Thomson Reuters in December 2011. The resulting shadow price of this study is also lower than that of other previous studies on other countries.⁸ For China, Wang et al. [31] use the directional distance function to estimate the CO₂ emissions shadow price for Chinese provinces in 2007, resulting a value of 475.3 Yuan per ton (about 73.1\$/T), this is much higher than the international CO₂ emissions trading market price, as well as the prices in other developed countries. According to the basic principle of environmental economics, the marginal abatement costs of emissions are negatively related with the amount of emissions and related positively with regulation levels. Additionally, China is well known to be the world's largest CO₂ emitter. Considering these two conceptual characteristics, the marginal abatement cost of emission in China should not be so high. This supports our results regarding the abundance of CO₂ emissions in China. Additionally, since CO₂ emissions are not officially regulated in China, the marginal abatement cost of CO₂ emissions may be to a low level. The reason for the high prices found in the study of [31] may derive from the use of directional output distance function which put the observations on a more steep portion of *T* when desirable outputs are expanded and undesirable outputs are contracted.

Regarding to province level, in the case of 2010, Zhejiang evidences the highest CO₂ emissions marginal abatement cost of 13.9\$/t, whereas Shanxi have the lowest values, of 3.13\$/t. Zhejiang had the highest CO₂ emissions efficiency score, indicating that Zhejiang is already using energy resources efficiently, and thus it will cost Beijing much more to abate additional CO₂ emissions. Shanxi is the largest coal production province and very high CO₂ emitting provinces. Therefore, Shanxi does not incur large expenses to abate an additional ton of CO₂ emissions from their already huge CO₂ emissions. Fig. 4 shows that the average marginal CO₂ abatement cost for China decreased from 2002 to

2004; from 2005, the cost begins to increase. Regarding the figure, the Chinese government began to regulate CO₂ and the abatement for CO₂ is becoming more difficult.

In the 12th Five-Year Plan (2011–2015), different reduction targets for CO₂ emissions for each province were assigned. When individual provinces are required to reduce their pollutant emissions by some amount, through emission trading with other provinces with lower marginal abatement costs, the province can achieve cost savings. Therefore, we argue that the Chinese government can enact a regional emission trade scheme (ETS) in the market system as each province is an administrative district. Through such an emissions trade scheme (ETS), a cost saving effect can be realized in regional economies. If the Chinese government brings the carbon emission trading between different provinces into the market, the market price of CO₂ emissions could be ranging from 3.13\$/t to 13.9\$/t corresponding to their marginal abatement costs in 2010, depending on the negotiation power of provinces [50].

In an effort to determine the relationship between CO₂ marginal abatement cost and CO₂ emissions efficiency, we employ a plotting method involving the Kernel density graphic and regression to observe these relations. We can show there to be a significant positive relationship between CO₂ marginal abatement cost and CO₂ emissions efficiency at the 1% level (Fig. 5). This suggests that regions with high CO₂ emissions efficiency will also suffer from high CO₂ marginal abatement costs. Therefore, governmental support for equitable market pricing on CO₂ marginal abatement costs or shadow prices should clearly be emphasized.

5. Conclusions

This paper employs a non-radial slacks-based measure (SBM) framework to analyze the reduction potential, efficiency, and abatement cost of CO₂ emissions in China. The results of our empirical study show that the average CO₂ emission reduction can reach levels as high as 56.1 million for each province and 1683 million tons nationwide. CO₂ emission efficiency varies from 0.146 to 1, with an average of 0.645. The economically well-developed east area evidences better CO₂ emission efficiency than the other two regions, whereas the west area, with the lowest level of economic development, evidences the lowest CO₂ emission efficiency. Thus, the equitable policy paradigm of carbon efficiency should be promoted only with great caution in order to avoid this dichotomy associated with eco-friendly economic development.

Generally speaking, the provinces of China did not perform carbon-efficiently. As China is currently in the initial stage of emission abatement, a lower market pricing system could be considered. The average marginal abatement cost of CO₂ emissions is about \$7.2. This result bears some important implications, encouraging policy-makers in the Chinese government to dispense with some of the many complicated regulations inherent to the current emission system, and to more actively promote international ETS transactions.

In any regard, challenging policies regarding the sustainable development by the Chinese government should be analyzed in more depth, as well as with far more implications. Our research is expected to shed certain light on the optimal path of economic growth with emphasis on the carbon emission efficiency and its attendant market system.

The limitation of this study is that the SBM DEA is a non-parametric method so one does not have to specify the function form. However, the production function is assumed to be the same for all provinces which can cause biased estimation. In the case of China with large provincial differences in the industry structure that assumption is relatively restrictive, therefore, this limitation should be noticed.

⁸ Rezek and Campbell [49] estimate the shadow prices of CO₂ for US coal-fired power plants with an average of \$18 per ton in 1998. Lee [37] employs plant-level data to estimate CO₂ marginal abatement costs in the Korean electricity generation sector in 2007, and calculated them to be an average of \$14.63 per ton.

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